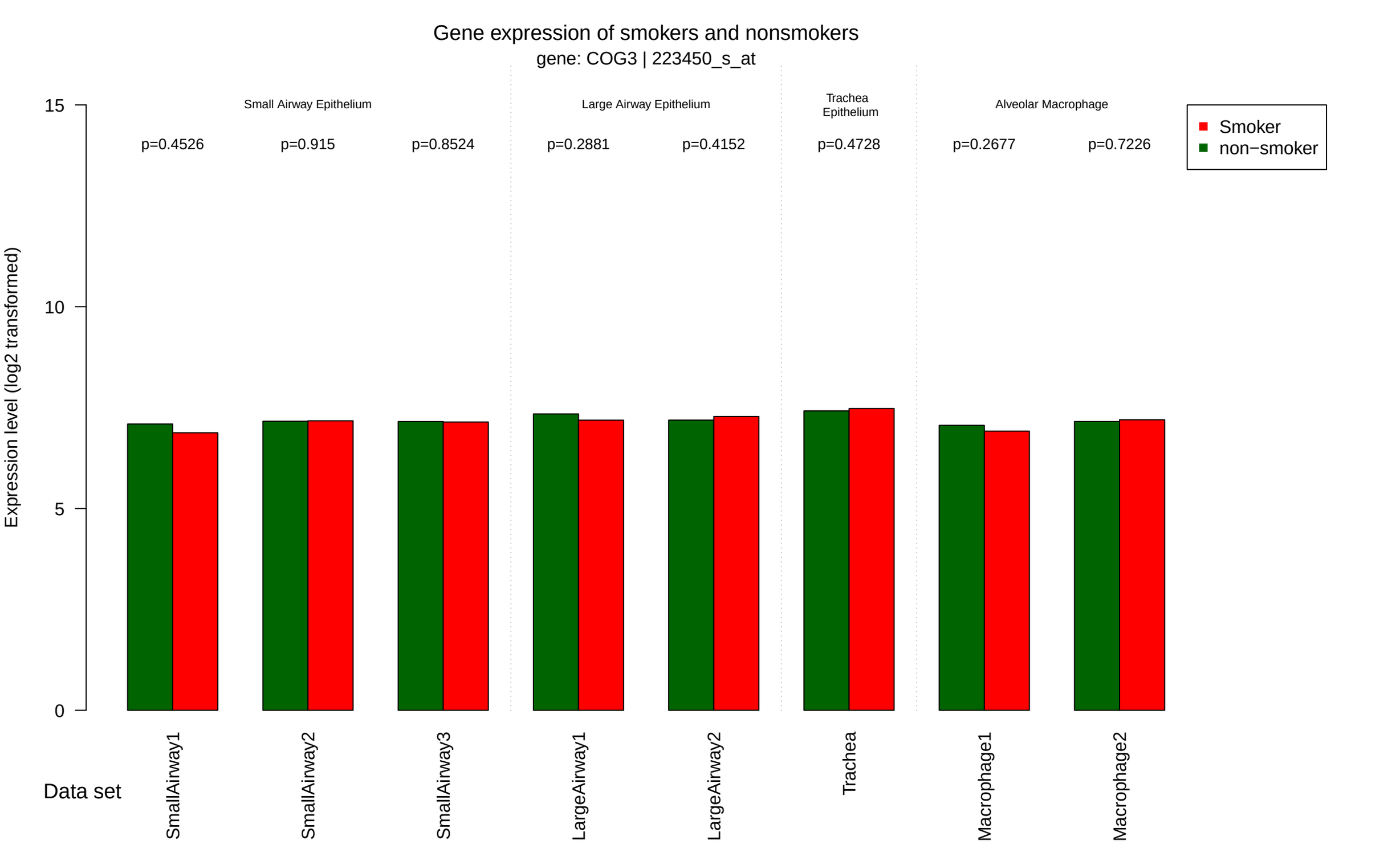
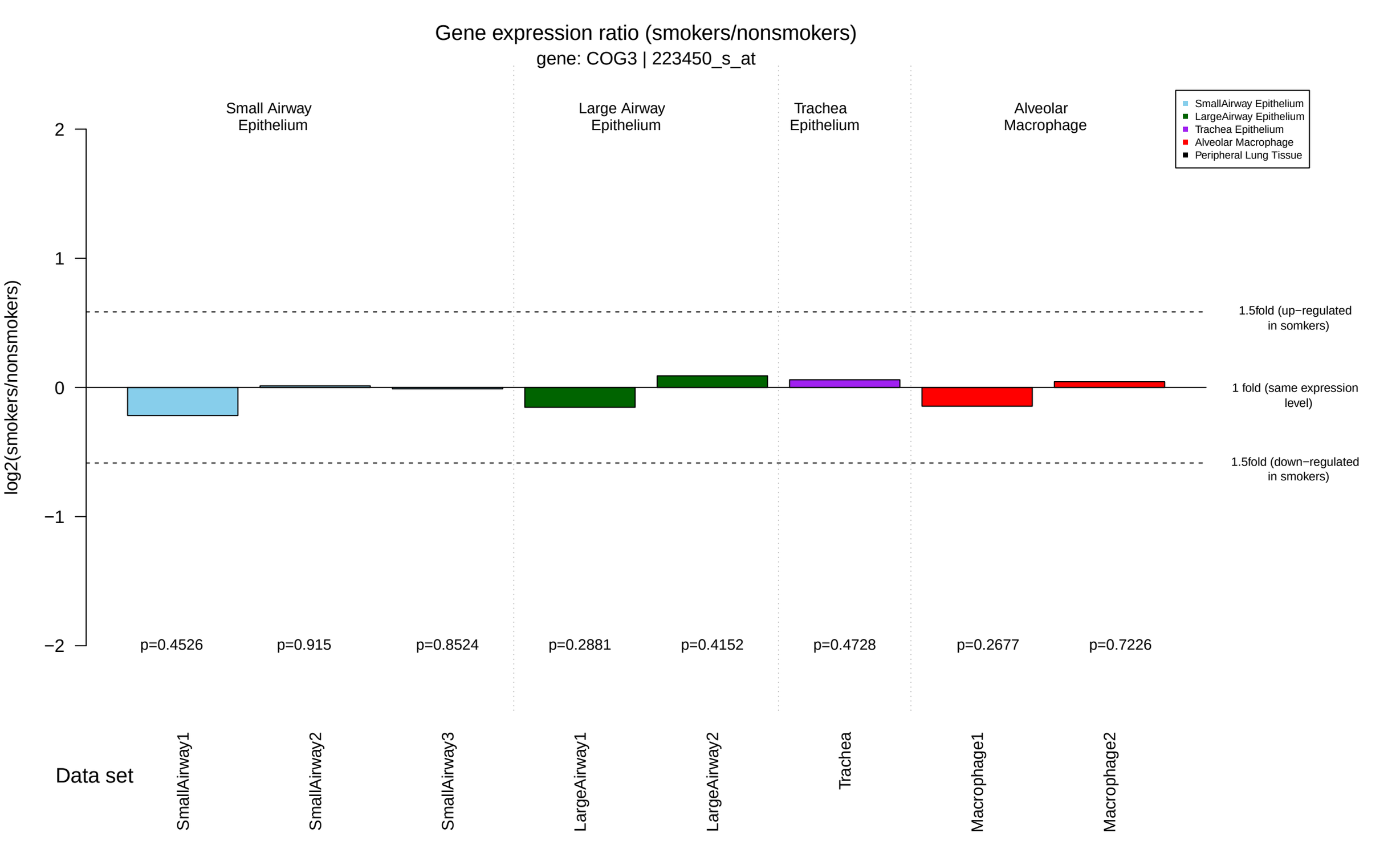
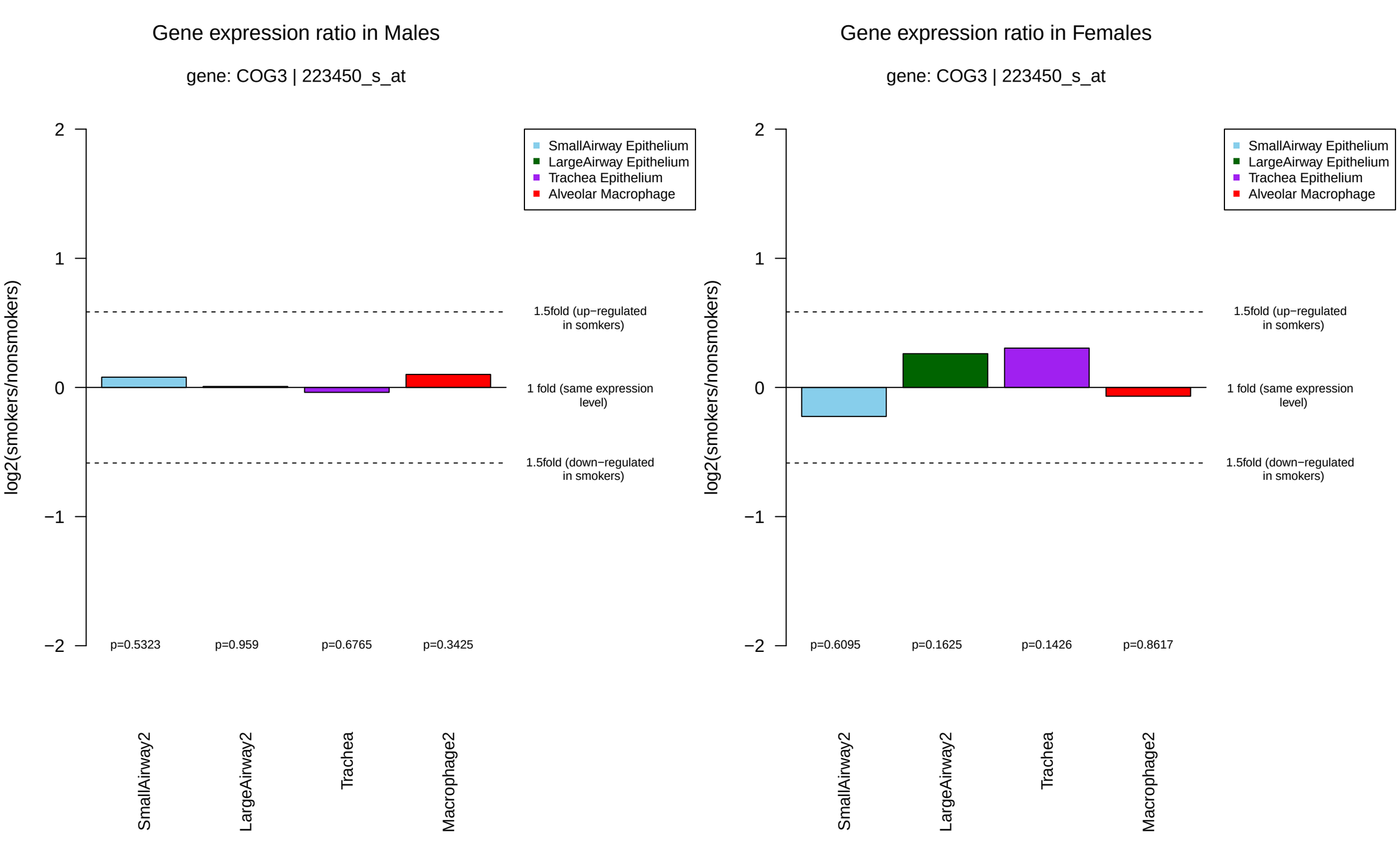
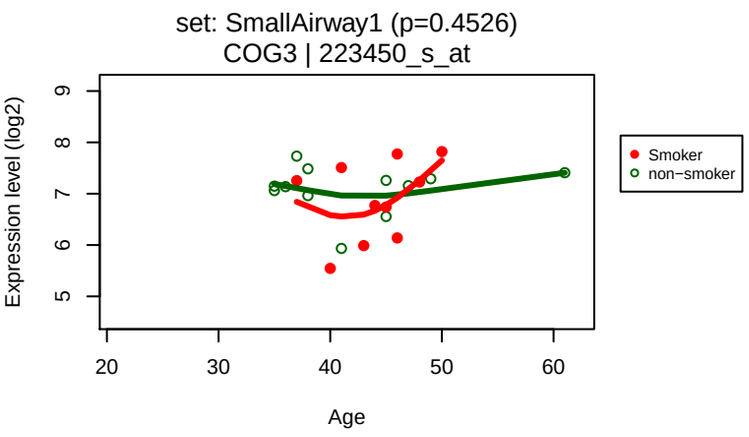
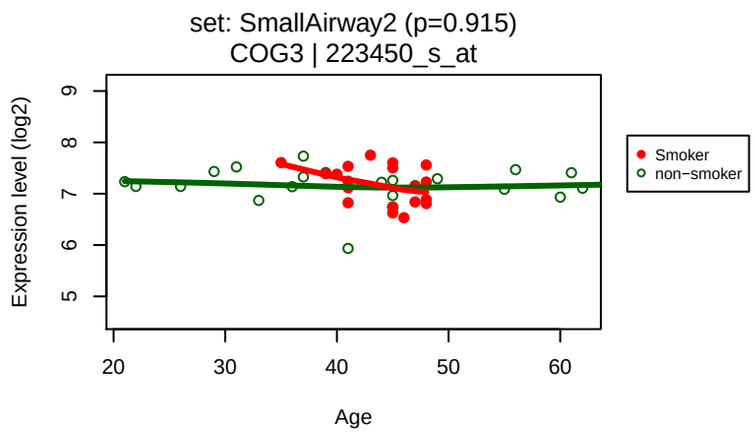
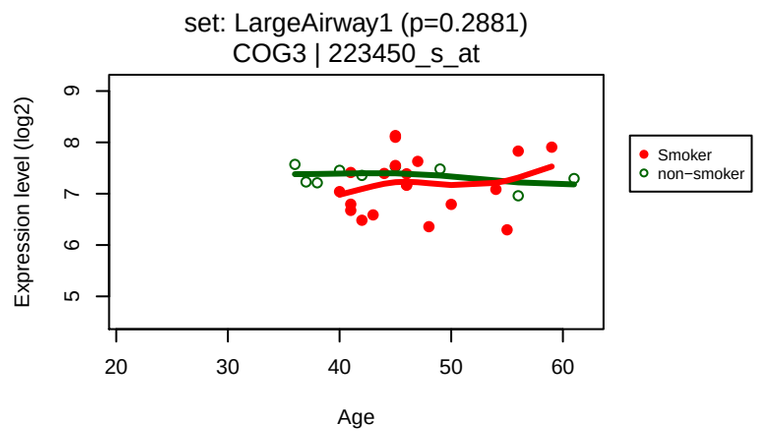

Gene expression of smokers and nonsmokers
gene: COG3 | 223450_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.4526
p=0.915
p=0.8524
p=0.2881
p=0.4152
p=0.4728
p=0.2677
p=0.7226
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

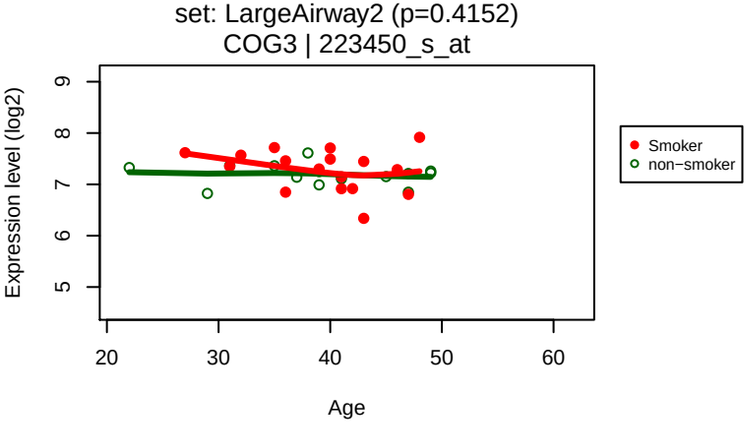

Gene expression ratio (smokers/nonsmokers)
gene: COG3 | 223450_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4526
p=0.915
p=0.8524
p=0.2881
p=0.4152
p=0.4728
p=0.2677
p=0.7226
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

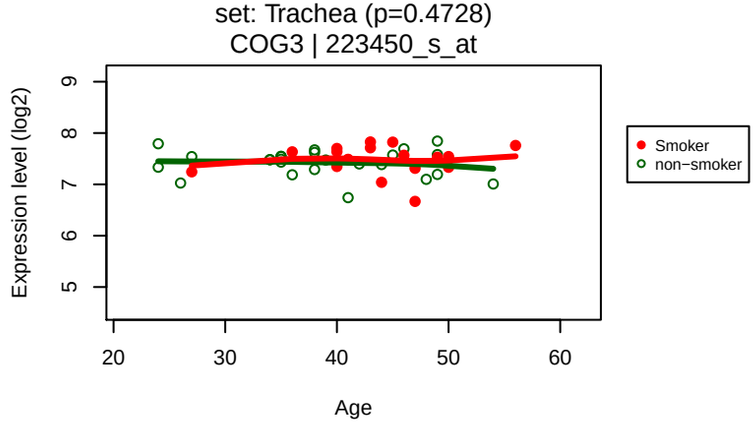

Gene expression ratio in Males
gene: COG3 | 223450_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5323
p=0.959
p=0.6765
p=0.3425
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: COG3 | 223450_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6095
p=0.1625
p=0.1426
p=0.8617
SmallAirway2
LargeAirway2
Trachea
Macrophage2

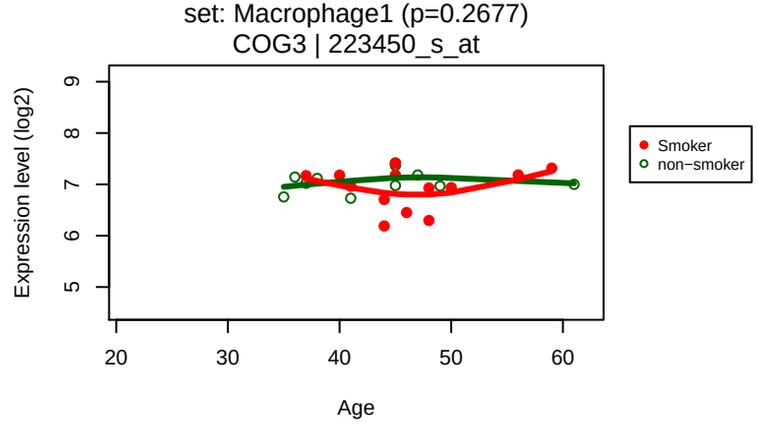

set: SmallAirway1 (p=0.4526)
COG3 | 223450_s_at
Expression level (log2)
Age
Smoker
non−smoker

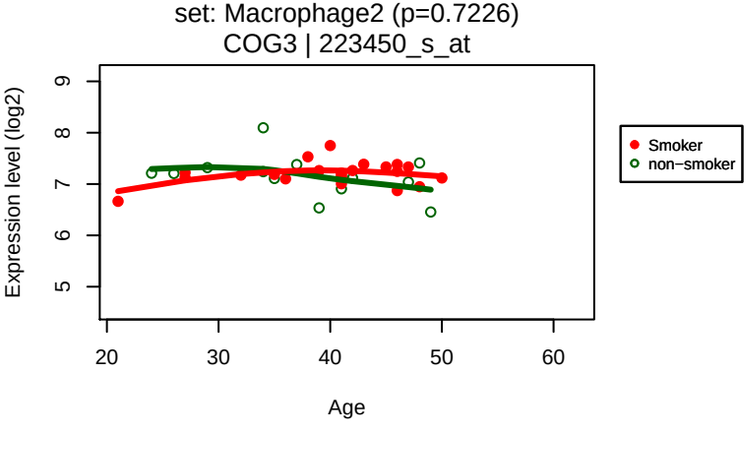

set: SmallAirway2 (p=0.915)
COG3 | 223450_s_at
Expression level (log2)
Age
Smoker
non−smoker

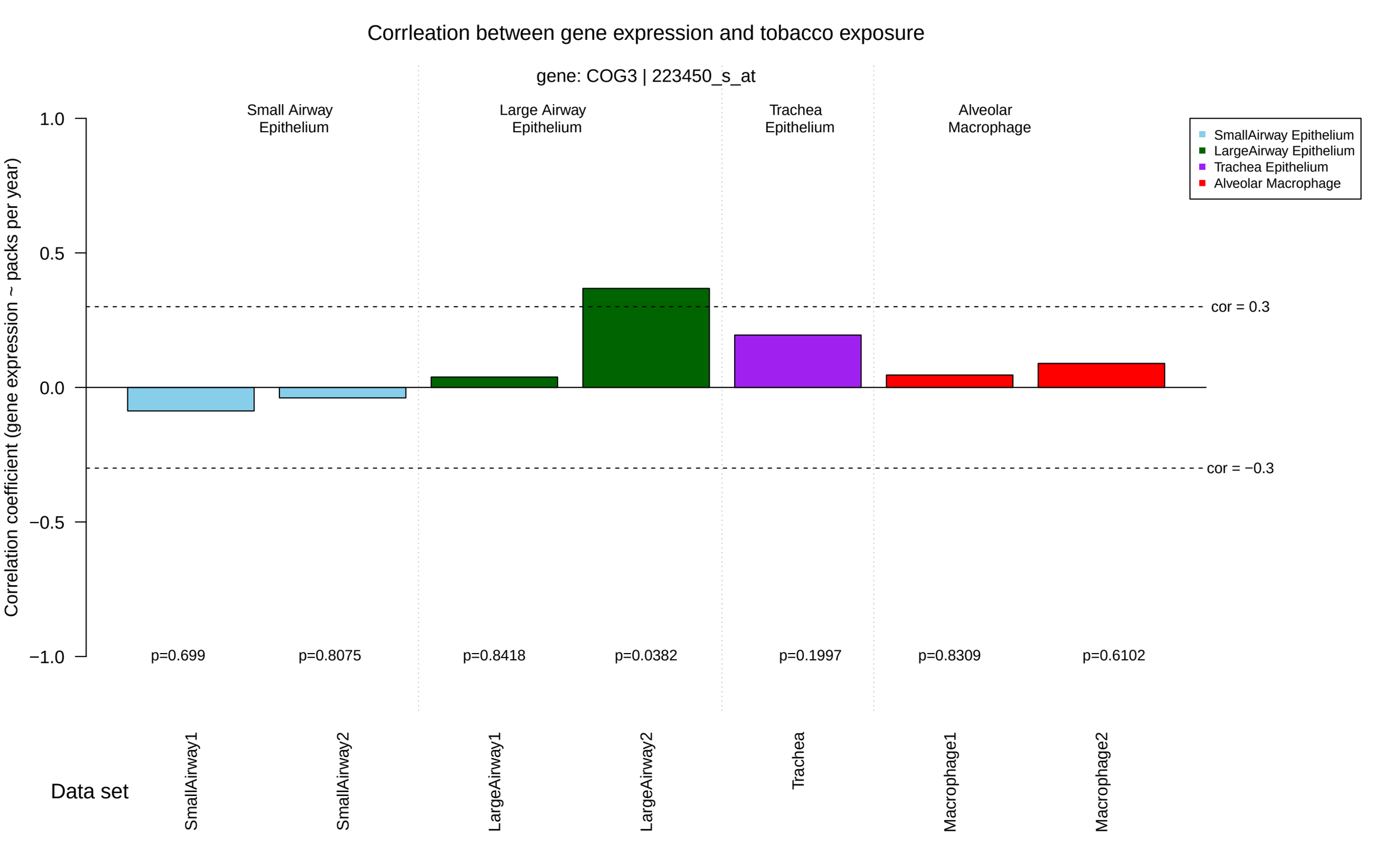

set: LargeAirway1 (p=0.2881)
COG3 | 223450_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.4152)
COG3 | 223450_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.4728)
COG3 | 223450_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.2677)
COG3 | 223450_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.7226)
COG3 | 223450_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: COG3 | 223450_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.699
p=0.8075
p=0.8418
p=0.0382
p=0.1997
p=0.8309
p=0.6102
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2